Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- Blockchain.com
- https://coinmarketcap.com/
- https://www.investopedia.com/

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure? Blockchain
3. Bitcoin uses which consensus mechanism? Proof-of-Work
4. The smallest unit of Bitcoin is called? Satoshi
5. Bitcoin is a peer-to-peer type of what system? Payment

Wallets

6. A wallet stores the user's Private key.
7. The recovery phrase used in wallets is called a Seed phrase.
8. Hardware wallets store keys Offline .
9. Bitcoin addresses are derived from Public keys.

10. What is required to sign a transaction? Private

Security

11. Bitcoin prevents double-spending using Blockchain .
12. Bitcoin is secured using Asymmetric cryptography.
13. To run a Bitcoin node, the software required is Bitcoin Core .
14. The process of verifying transactions is called Mining .
15. The Bitcoin network is described as Decentralized.

Mining

16. Miners solve Cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created Bitcoins .
18. The block time target for Bitcoin is 10 minutes.
19. Mining difficulty adjusts every 2016 block.
20. Miners compete to find a valid block Hash.

Blockchain Structure

21. Each block contains a Merkle root.
22. Blocks are linked using the previous block's Hash .
23. A Bitcoin transaction contains inputs and outputs .
24. Unspent transaction outputs are called UTXOs .
25. The first-ever Bitcoin block is called the Genesis block.

Network

26. A full node stores the entire blockchain Ledger .

27. Bitcoin uses which protocol for communication? P2P

28. A light wallet uses SPV mode (SPV/full).

29. Nodes relay information across a Peer-to-peer network.

30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the latest block:

- Block height: 961,126
  Block hash: 00000000000000000001473ebce7be38e82f9c332a0cafdbed00310db15e7ef9
- Timestamp:Aug 05, 2026, 12:33:42 PM
- Number of transactions: 4,175
- Miner/pool name: Unknown

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction :
    7cc664d96e7aae2c9db21aa529c41e62f54de5e690da6b8899afd429687077ea.

11. Number of inputs: 1

12. Number of outputs: 2

13. Transaction fee: 0 SATS

14. How many confirmations does it have: 1

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block:
    bc1qukghnu65atvwte8rhsfwa9m7c5chq5qr08x7sk

16. Record:

- Total BTC received: 7.99944728 BTC
- Total BTC sent: 6.89967006 BTC
- Current balance: 1.09977722
  What is the role of a full node? A full node stores the complete blockchain, verifies all transactions and blocks independently, and helps secure and decentralize the Bitcoin network.

List any two developer resources available on bitcoin.org.

- Documentaries
- Podcasts

Explore : https://coinmarketcap.com/

Task C1: Market Information

- Current price of BTC: ~$64,292.90 USD
- 24-hour % change: +0.76%
- Market capitalization: ~$1.28 trillion USD
- Circulating supply: ~20.06 million BTC
- Total/maximum supply: 21.00 million BTC

Task C2: Historical Data

- Price one year ago (August 2025): Trading around $115,000–$117,000 USD (prior to peaking near $126,000+ later in the fall).
- 30-day high (July–August 2026): ~$65,326.16 USD
- 30-day low (July–August 2026): ~$57,832.50 USD

Explore https://www.investopedia.com/

Why is Bitcoin considered volatile?

Bitcoin's price is highly volatile because it is driven by shifting supply and demand dynamics, intense media coverage, celebrity influencer opinions, and changing government regulations. Unlike traditional financial assets, its value is deeply affected by speculative investor sentiment and the actions of "whales" who hold massive positions

List two investment risks mentioned.

Regulatory Risk: Evolving global laws, taxes, or government crackdowns can negatively impact its market value and usability.

Security & Capital Loss Risk: Breaches on un-backed exchanges, wallet hacks, or losing your wallet's private keys results in a permanent and irreversible loss of funds

🔹 Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Investopedia provides a far clearer understanding of cryptocurrency than raw data tracking sites like CoinMarketCap. While CoinMarketCap is excellent for scanning real-time numerical figures, Investopedia explicitly unpacks the underlying economic and technical drivers behind market performance. It educates investors on macro risks and technical definitions rather than just presenting fluctuating price points.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain establishes trust without a central authority by utilizing a decentralized, public ledger that copies and spreads data across a global network of independent computer nodes. The entire ecosystem operates according to hardcoded software rules where new transactions must be validated via algorithmic consensus mechanisms—such as Proof-of-Work—requiring network agreement before data can be saved. Once recorded, the shared recordkeeping structure mirrors the information perfectly across thousands of nodes. This redundancy makes the ledger completely immutable and secure, rendering fraudulent database tampering mathematically impossible without triggering immediate rejection by the rest of the network.